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# HALO REACH: Project Coordination through Systems Engineering, Risk Analysis and Conceptual Design

UAS for Disaster Relief - AE3 2024

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## **Abstract**

This report details the project coordination side of a disaster relief and communication relay UAS, looking at the aspect of conceptual design, systems engineering and an in-depth analysis of risk. The report initially explores the steps taken in obtaining a conceptual design for the UAS through teamwork and coordination, allowing subteams to then focus on preliminary sizing and detailed design. A SWOT analysis identified internal strengths, weaknesses, opportunities, and threats. It helped leverage strengths to achieve goals, address weaknesses to improve performance, capitalize on growth opportunities, and mitigate potential threats. A V-model was used to to emphasise verification and validation across the conceptual design phase. It allowed for any early risk or defect detection in the case of following the full life cycle of the product. An N-Squared diagram displayed the interactions and interfaces between all system components across the lifecycle. A House of Quality was generated in order to assess our competitors, as well as translating customer requirements into technical requirements and assessing their importance with regards to direct competitors in the market. An in-depth risk analysis details how one can discover and allocate relevant risks in the holistic view of the project, how they will affect the project as a whole and how to mitigate against them.

# Contents

|                                                          |            |
|----------------------------------------------------------|------------|
| <b>Table of Contents</b>                                 | <b>ii</b>  |
| <b>List of Figures</b>                                   | <b>iii</b> |
| <b>List of Tables</b>                                    | <b>iv</b>  |
| <b>Nomenclature</b>                                      | <b>v</b>   |
| <b>1 Introduction</b>                                    | <b>1</b>   |
| <b>2 Project Management</b>                              | <b>1</b>   |
| 2.1 Requirements . . . . .                               | 1          |
| 2.2 Conceptual Design . . . . .                          | 1          |
| 2.2.1 Configuration . . . . .                            | 1          |
| 2.2.2 Conceptual sizing . . . . .                        | 2          |
| 2.3 Regulations . . . . .                                | 3          |
| 2.3.1 Choice of regulations . . . . .                    | 3          |
| 2.3.2 Constraining factors from regulations . . . . .    | 3          |
| <b>3 Systems Level Engineering</b>                       | <b>4</b>   |
| 3.1 V-model . . . . .                                    | 4          |
| 3.2 SWOT analysis . . . . .                              | 5          |
| 3.3 N <sup>2</sup> chart . . . . .                       | 5          |
| 3.4 House of Quality . . . . .                           | 5          |
| 3.4.1 Primary competitors . . . . .                      | 6          |
| <b>4 Risk analysis</b>                                   | <b>7</b>   |
| 4.1 What is risk? . . . . .                              | 7          |
| 4.2 Risk map . . . . .                                   | 7          |
| 4.3 Optimising risk analysis . . . . .                   | 7          |
| 4.4 Risk categories . . . . .                            | 8          |
| 4.4.1 Regulatory compliance . . . . .                    | 8          |
| 4.5 Human factors . . . . .                              | 8          |
| 4.6 Technical . . . . .                                  | 9          |
| 4.7 Environmental . . . . .                              | 10         |
| 4.8 Operational . . . . .                                | 10         |
| 4.9 Cost . . . . .                                       | 11         |
| 4.10 Risk management and mitigation strategies . . . . . | 11         |
| <b>5 Conclusion</b>                                      | <b>12</b>  |
| <b>References</b>                                        | <b>13</b>  |
| <b>A N<sup>2</sup> chart</b>                             | <b>16</b>  |
| <b>B SWOT analysis</b>                                   | <b>17</b>  |
| <b>C House of Quality</b>                                | <b>18</b>  |
| <b>D Complete risk table</b>                             | <b>19</b>  |
| <b>E Mitigation techniques</b>                           | <b>20</b>  |

**List of Figures**

3 Pugh matrix depicting final conceptual choice for REACH . . . . . 2

4 Illustration of a conceptual V-Model which can be used throughout the UAS life cycle. 4

5 Caption . . . . . 16

6 SWOT analysis performed for HALO REACH . . . . . 17

7 House of Quality analysis of REACH . . . . . 18

8 Complete risk table including all details for most important detailed risks for each section 19

List of Tables

|   |                                                                                           |    |
|---|-------------------------------------------------------------------------------------------|----|
| 1 | People involved in HALO REACH - GDP ( <b>student in bold</b> ) . . . . .                  | 1  |
| 2 | Comparison of UAV Types Based on Subteam Rating . . . . .                                 | 2  |
| 3 | List of most similar competitors in the large scale drone industry and comparative traits | 6  |
| 4 | List of relevant mitigation techniques for most intense risks . . . . .                   | 20 |

## Nomenclature

|                    |                                                                    |
|--------------------|--------------------------------------------------------------------|
| $\ddot{h}$         | Acceleration in y-direction (m/s <sup>2</sup> )                    |
| $m_{\text{total}}$ | Total mass (kg)                                                    |
| $N$                | Normal force (N)                                                   |
| $P$                | Power (W)                                                          |
| $x_T$              | Thrust moment arm (m)                                              |
| CAA                | Civil Aviation Authority                                           |
| CAD                | Computer Aided Design                                              |
| CFD                | Computational Fluid Dynamics                                       |
| CFR                | Code of Federal Regulations                                        |
| EASA               | European Union Aviation Safety Agency                              |
| FAA                | Federal Aviation Administration                                    |
| FCC                | Flight Control Computer                                            |
| GCS                | Ground Control System                                              |
| HALE               | High Altitude Long Endurance                                       |
| HALO               | Humanitarian Aid Logistics Operations                              |
| HoQ                | House of Quality                                                   |
| ICAO               | International Civil Aviation Organisation                          |
| LQR                | Linear Quadratic Regulator                                         |
| MALE               | Medium Altitude Long Endurance                                     |
| MI                 | Merit Index                                                        |
| MPC                | Model Predictive Controller                                        |
| MTOW               | Maximum Takeoff Weight                                             |
| OCHA               | United Nations Office for the Coordination of Humanitarian Affairs |
| PIDF               | Proportional, Integral, Derivative, and Filter                     |
| REACH              | Remote Emergency Assistance and Communication Hub                  |
| RMS                | Root Mean Square                                                   |
| SWOT               | Strengths Weaknesses Opportunities Threats                         |
| UAS                | Uncrewed Aerial System                                             |
| UAV                | Unmanned Aerial Vehicle                                            |
| VTOL               | Vertical Take-Off and Landing                                      |



# 1 Introduction

Across the globe, natural disasters can strike without warning. Earthquakes, hurricanes, floods, and other catastrophic events often devastate communities, leaving them isolated and in dire need of assistance. In many cases, relief efforts are hampered by damaged infrastructure, adverse weather conditions, and communication breakdowns. Conventional aircraft and ground vehicles struggle to navigate the treacherous terrain and blocked pathways, making it difficult to deliver essential supplies and medical aid. This results in more casualties, larger-scale devastation, and prolonged suffering for those affected [1]. This international issue must be addressed with innovative solutions to mitigate further risks and enhance the effectiveness of disaster response efforts.

Here at HALO REACH, our aim is to develop a versatile, modular VTOL UAS capable of reaching regions that others can't, to provide prompt disaster relief or act as a communications relay for those on the ground. The UAS will conduct the mission by taking-off and landing vertically from ship decks under adverse conditions, cruising to and from the site of a disaster [reference project brief].

The team is comprised of 24 members in 6 sub teams, where each subteam is assigned an academic supervisor with expertise in the respective subteam field. The project coordination team deals with the overall project management, systems engineering, budgeting, as well as organising weekly presentations for all members and academic supervisors to discuss and improve the project. The name 'HALO REACH' is an acronym for 'Humanitarian Aid Logistics Operations - Remote Emergency Assistance and Communications Hub', which reflect the core mission and capabilities of our UAV system. The acronym underscores our focus on logistics operations that are agile, reliable, and capable of overcoming the challenges posed by damaged infrastructure and adverse conditions, as well as providing swift and effective humanitarian aid in disaster-stricken areas, ensuring that critical supplies and communication support reach those in need without delay.

This report will be structured to primarily discuss the conceptual design of the drone, the systems engineering aspects of our company and the overall risk analysis.

## 2 Project Management

### 2.1 Requirements

As project coordinators, it is key to underline the baseline requirements of the mission at hand. Guidelines given to the team stipulate that the drone has to be a versatile, modular VTOL UAS, and it must either:

- Serve as a long endurance communications relay
- Land vertically to deliver relief supplies

REACH must be able to conduct the mission by taking off and landing vertically from ship decks under adverse conditions on a 6x6 metre platform, cruising to and from the site of a disaster. It must be able to carry either a 20kg payload for the long endurance mission, or a 50kg payload for the supply delivery, resulting in an MTOW of 160kg. REACH must also be able to cover a 185km radius for its missions [2]

### 2.2 Conceptual Design

#### 2.2.1 Configuration

The conceptual design process involves investigating different topologies. These designs were researched until a collection of 9 were finalised. A ranking matrix was then generated based on the pros and cons of each topologies per each subteam, as shown in Table 2

|                              | Vector-<br>ed<br>thrust | Tilt<br>wing | Wing-<br>less | MALE-<br>/HALE | VTOL<br>fixed<br>wing | Quad-<br>/hex-<br>copter | Single<br>rotor | Coaxial | Blimp |
|------------------------------|-------------------------|--------------|---------------|----------------|-----------------------|--------------------------|-----------------|---------|-------|
| Aerodynamics                 | 8                       | 2            | 1             | 10             | 7                     | 5                        | 4               | 3       | 6     |
| Flight<br>Dynamics           | 8                       | 6            | 6             | 5              | 7                     | 7                        | 7               | 7       | 3     |
| Modular<br>add-on<br>systems | 7                       | 6            | 3             | 5              | 8                     | 3                        | 6               | 5       | 5     |
| Structural<br>design         | 4                       | 8            | 10            | 5              | 7                     | 8                        | 5               | 7       | 7     |
| Systems and<br>Propulsion    | 10                      | 6            | 5             | 9              | 10                    | 7                        | 5               | 4       | 10    |

**Table 2.** Comparison of UAV Types Based on Subteam Rating

Many exotic varieties of UAV's were explored, ranging from wingless drones, quadcopters, blimps and fixed-wing VTOL UAV's. A scoring systems from 1 to 10 was used, where 10 was the best score. Three configurations were selected from the subteam scoring matrix, where these three were then re-evaluated and scored on additional manufacturing and engineering criteria, alongside a relevant weighting multiplier for each score in a new Pugh matrix [3]. These multiplier was decided with all subteams to ensure consistency. All teams then democratically voted on the final ranking of each configuration with respect to the criteria. This Pugh matrix can be seen in 3, where the scoring is between 1 to 3, representing the worst to best configurations for the respective engineering criteria.

| Criteria                 | Weighting | Fixed wing UAV VTOL | Tilted/folding wing | Wingless (Helicopter style) |
|--------------------------|-----------|---------------------|---------------------|-----------------------------|
| Endurance                | 10        | 2                   | 3                   | 1                           |
| Aircraft weight          | 5         | 2                   | 1                   | 3                           |
| Packaging                | 2         | 1                   | 3                   | 2                           |
| Design complexity        | 6         | 3                   | 1                   | 2                           |
| Manufacturing complexity | 6         | 3                   | 1                   | 2                           |
| Cost                     | 7         | 2                   | 1                   | 3                           |
| Innovation               | 3         | 2                   | 3                   | 1                           |
| Environment/Robustness   | 8         | 3                   | 2                   | 1                           |
| Controllability          | 8         | 2                   | 1                   | 3                           |
| Overall                  | -         | 128                 | 93                  | 109                         |

**Figure 3.** Pugh matrix depicting final conceptual choice for REACH

The final scoring displays that a fixed wing VTOL UAV would be the most sensible for our mission profiles, heavily based on its robustness in different terrain environments, its design complexity and its manufacturing complexity [4].

Different configurations of fixed wing VTOL UAV's were explored in order to gauge possible design ideas and concepts. The sheer volume of different design choices, even when limited in configuration, resulted in many ideas for different components and sub-components of the vehicle. Major design choices that were decided together as a team were the number of rotors, propeller layout, tail configuration and undercarriage choice. This is due to several teams being affected by the choice of these parameters.

### 2.2.2 Conceptual sizing

Once the final configuration was selected, the subteams were then allocated their respective roles for conceptual sizing. These are illustrated below:

| Aerodynamics               | Structures               | Modular add-ons                     |
|----------------------------|--------------------------|-------------------------------------|
| Wing design                | Aircraft component loads | Secondary market add-on analysis    |
| Rotor and propellor design | Material selection       | Structural analysis of add-ons      |
| Winglet design             | CAD                      | CAD of payload and add-ons          |
| CFD                        | Structural analysis      | Payload attachment configuration    |
| Lift and drag analysis     | Manufacturer methods     | Payload attachment mechanism design |

| Flight Dynamics                                 | Systems and Propulsion                            |
|-------------------------------------------------|---------------------------------------------------|
| State space and stability analysis              | Fuel and propulsion selection                     |
| Takeoff and landing controllers                 | Weight and balance                                |
| Fixed wing longitudinal and lateral controllers | Navigation and sensor selection                   |
| Transition controller between flight modes      | Communication and data relay                      |
| High fidelity simulation environment            | System architecture, FCC, sensor fusion framework |
| Robustness studies                              | GCS and Launch and Recovery architecture          |

## 2.3 Regulations

### 2.3.1 Choice of regulations

Once the configuration of the drone was selected, it is key to explore which regulations would apply to its classification. In order for our drone to operate across the globe, it must adhere to several international regulations. These regulations are in place to ensure the safety, legality and successful operation of any UAV [5].

Our drone qualifies under the **Certified Category** of UAV's [6], as due to its weight and operation it would pose a similar level of threat as a manned vehicle. As there are currently no official documents covering drones of this genre, we will have to adhere to the same regulatory regime as manned aircraft. By designing our UAV in compliance with these standards, we aim to maximize its reach and effectiveness in disaster relief efforts [7].

The three primary international regulatory bodies are the **International Civil Aviation Organisation (ICAO)**, the **European Union Aviation Safety Agency (EASA)** and the **Federal Aviation Administration (FAA)** [8] [9]. The two primary international humanitarian law and aid regulatory bodies include the **United Nations Office for the Coordination of Humanitarian Affairs (OCHA)** [10] and the **Red Cross and Red Crescent Societies**. With regard to the former group, these provide several annexes which detail the required the operation, airworthiness and safety management of any UAV operating globally.

### 2.3.2 Constraining factors from regulations

As REACH contains aspects of VTOL and fixed wing flight, manned aircraft regulations for light aircraft, as well as helicopters, were used in order to establish key constraints that are not underlined in the original briefing.

**Bank angle** for fixed wing aircraft is defined in the EASA and FAA guidelines as "limited to 30 degrees for normal operations". As the aircraft will not perform any aerobatic operations, this will be our bank limit [11][12].

**Climb rate** for light aircraft is defined in the EASA and FAA regulations as "at least 300 feet per minute at maximum takeoff weight", which is used in any sizing constraints in aerodynamic performance and analysis [13][12].

**Collisions with foreign bodies** is defined under the EASA and FAA regulations; for bird strikes the UAV must be designed to withstand the impact of a "1.8kg bird at design cruise speed." [14].

This number is increased for the empennage structures for the UAV's maximum cruise speed. With regards to other foreign bodies, the operators will need to take into account the risks associated with operating REACH in its designated environment, whether that be onboard a ship for the primary missions, or performing in any of the secondary markets. [15].

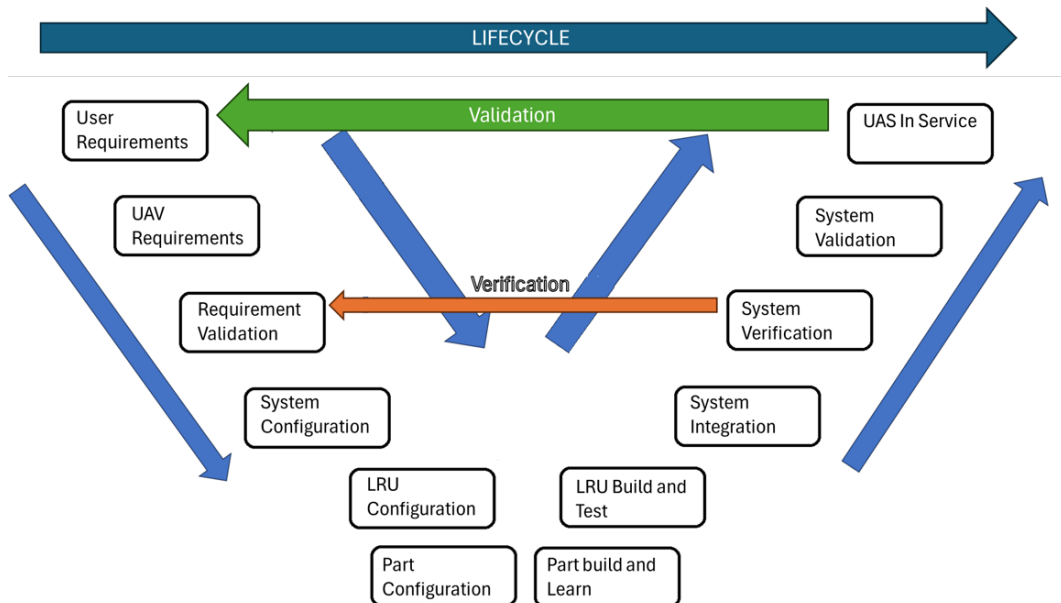
### 3 Systems Level Engineering

Systems engineering is an "interdisciplinary approach and means to enable the realisation of successful systems" [16]. As the UAV is comprised of numerous components, there will be a large number of interacting subsystems onboard the UAV which will only be properly integrated amongst each other to provide the most optimal performance of the UAV by employing systems engineering approaches.

#### 3.1 V-model

A design structure is essential to implement early when working on a large-scale project with a team of this caliber. A well designed structure allows for manageable tasks and components to be allocated to relevant parties. This allows for complexity to be broken down and further simplified, crating a much more achievable workflow. It can also allow for a paralel-style working environment, improving overall efficiency across the lifecycle of the system. This is illustrated in greater detail in Patricia's report by the use of team and subteam Gantt charts [17].

Once a design structure has been implemented, there needs to be some sense of direction with regards to the overall workflow of the project. This is where the V-Model comes in. The V-Model is effectively a waterfall model used to depict the overall lifecycle, whilst implementing **Verification** and **Validation** as the process are carried out. Verification is the process of deeming whether a system has been conceptualised and designed in accordance to the requirements, whilst validation is the process of assessing whether a system outputs the intended results [18]. The V-Model addresses potential risks and flaws in the system, which allows for early detection and mitigation. In the case of HALO REACH, we adapt the V-Model to underline our requirements and serve as a core for all design processes undergone by the teams. It can be seen in 8



**Figure 4.** Illustration of a conceptual V-Model which can be used throughout the UAS life cycle.

### 3.2 SWOT analysis

When it comes to planning the complete conceptual design process, it's always good to be aware of several key factors of your target industry and your product. A SWOT analysis performs exactly this; it provides one with a holistic overview of all the key factors that can affect the system and/or company at any times along the lifecycle:

- **Strengths** and **Opportunities** of a system can help the company display the advantages and leverage the drone will have over other competitors, allowing one to climb through the market and position the drone as the market leader.
- **Weaknesses** and **Threats** are crucial to underline within the company, as well as the product. This permits for further strategisation and improvement to the system, including the mitigation of any possible risks or challenges. This could be internal within the company, the physical drone or the environment in which it operates in.

After underlining all the aspects of a SWOT analysis, we can allocate any further research towards topics and systems that may need more attention. This included enhancing the payload capacity mechanism, improvements in modularity and improvements in operability in all conditions. These regions are some of the more demanding sectors within the UAS. However, if these are nailed down, the potential for the drone to flourish in its primary and secondary markets will naturally surge. The SWOT analysis is underlined below in Appendix B

### 3.3 N<sup>2</sup> chart

There are several methods that one can use to portray the data flow within a system, from introduction of an idea, production, testing and its demise. The N<sup>2</sup> chart allows for visualise the interactions and interdependencies between different subsystems or processes within the overall system. This simplifies the understanding of the system architecture, allowing for tasks to be split accordingly and worked on simultaneously [19]. Interacting subsystems can be contributed to accordingly, increasing efficiency of the design process. Integration of subsystems would also be simplified from the integration of the N<sup>2</sup> chart. The full chart can be seen in Appendix A

### 3.4 House of Quality

The House of Quality (HoQ) is a powerful tool in establishing the relationship between customer requirements and engineering specifications [20]. It ensures that customer needs are understood and underlined, by the use of a customer importance rating for each requirement. A relationship value is also assigned to each engineering specification with respect to the customer requirement being assessed. The values were 9 for a strong relationship, 3 for a medium relationship and 1 for a weak or no relationship. The importance rating is then calculated with the following equation:

$$\text{Importance Rating of TR}_i = \sum_{j=1}^n (\text{Importance of CR}_j \times \text{Relationship Value}_{ij})$$

Where:

- **Importance Rating of TR<sub>i</sub>** is the importance rating of the i-th engineering specification
- **Importance of CR<sub>j</sub>** is the importance of the j-th customer requirement
- **Relationship Value<sub>ij</sub>** is the relationship rating between the j-th customer requirement and the i-th technical requirement.
- **n** is the number of customer requirements.

Once these are compared to the engineering constraints, it becomes trivial to allocate any resources towards the research, development and design of the key customer prerequisites. These qualities are also compared to significant competitors. By analysing the competitor’s strength and weaknesses, combined with our importance rating, the company can set targets, which will enable a superior drone to be produced, outperforming the competition.

The HoQ for HALO REACH can be seen in Appendix C. The key importance ratings underlined by the analysis were endurance and modular design. This highlights the importance of designing a drone with long endurance capabilities, as well as one that is operable in all conditions and scenarios.

### 3.4.1 Primary competitors

As this market is relatively novel, there are few drone manufacturers with UAV’s with similar requirements or mission profiles to HALO REACH. However, several drones were identified.

- **SMD** is one of China’s leading drone manufactueres. Their **V800** is their latest large scale hybrid transport UAV, primarily designed to carry up to 50kg of emergency materials. It also possesses 10 hours of endurance, flying 4.5hrs on its maximum payload capacity. It also has different configuration options, namely a visible light camera, infrared thermal imagery and LIDAR [21][22].
- **JOUAV** is, according to markets, China’s biggest large scale UAV manufacturer for industrial applications. Their **CW-80E** is their best drone for long endurance, long range , high payload capacity missions. Their drone is currently in use across China by companies like COMAC (aircraft manufacturer), as well as South Korean and Malaysian aerial research companies [23][24].
- The European market is relatively isolated, with only a handful of competitors focusing towards the military-end of UAV operation. **ElevonX** designs medium to large scale advanced UAV’s. Their fixed wing drones have been adapted to include VTOL capabilities, allowing them to target a broader market, similar to HALO REACH. Their **SkySierra** is the most similar UAV to REACH [25][26].

The table below depicts a comparison of all the listed competitors with REACH

**Table 3.** List of most similar competitors in the large scale drone industry and comparative traits

| Name      | Endurance (hours) | Maximum Takeoff Weight (kg) | Maximum Payload (kg) | Dimensions (m)  | Cost (\$) |
|-----------|-------------------|-----------------------------|----------------------|-----------------|-----------|
| REACH     | 10                | 160                         | 50                   | 6.0 x 3.8 x 2.3 | 460,000   |
| V800      | 10                | 230                         | 50                   | 8.2 x 3.7 x 1.0 | 465,000   |
| CW-80E    | 8                 | 100                         | 20                   | 5.0 x 3.1 x 3.8 | 525,000   |
| SkySierra | 5                 | 12.5                        | 3                    | 3.2 x 1.5 x 1.5 | 345,000   |
| Tango     | 6                 | 20                          | 5                    | 3.0 x 1.9 x 1.5 | 293,400   |

## 4 Risk analysis

### 4.1 What is risk?

Risk is a guaranteed aspect that all companies have to navigate. It is essentially the frequency at which an individual is expected to sustain any given level of harm from the realisation of specific hazards. [27] They can arise from several locations, namely **Technical**, **Schedule** or **Cost** associated risks:

- **Technical risks** arise from being unable to satisfy the product's stated requirements. This may be due to misunderstandings between teams or a lack of interaction between subcomponents.
- **Schedule risks** Come from the product not satisfying it's relevant requirements on time. This can occur during any stage of development, hence it is key to adhere to a set schedule and have mitigations in place.
- **Cost risks** are a consequence of the program exceeding the initial allocated budget. This is unfortunately common in the aircraft and drone industry, ranging from the Boeing 787 which went \$12-18 million over budget [28] to the RQ-4 Global Hawk, seeing costs rise substantially from \$10 million in 1994 to \$60.9 million by 2001 to a massive \$131.4 million by 2013 [29].

### 4.2 Risk map

The official definition of risk is the product of the likelihood and the impact. Risk is inherently difficult to quantify, as the scoring of the likelihood and impact of any named risk varies between engineers. Experience and judgement is key to estimate 'relative risk' levels. [source lecture 8].

A risk map can make it easy to quantify the likelihood and impact of risks that can arise in the lifecycle of the system:

- The **likelihood** is scored from A to E, where A is the lowest likelihood and E being a near certain likelihood. These levels are based on previous examples and evidence with regards to each risk and teh context around the system. However, for certain sectors it is still difficult to accurately quantify and justify some likelihoods.
- The **impact** is scored from 1 to 5, where 1 is the lowest and 5 is the largest impact. This can affect the system goals and the shceduling of the system, with a direct relationship to the cost factor of the subsystem the risk is allocated under.

|            |   | Risk Severity |       |       |           |              |
|------------|---|---------------|-------|-------|-----------|--------------|
|            |   | Negligible    | Minor | Major | Hazardous | Catastrophic |
| Likelihood | E |               |       |       |           |              |
|            | D |               |       |       |           |              |
|            | C |               |       |       |           |              |
|            | B |               |       |       |           |              |
|            | A |               |       |       |           |              |
|            |   | 1             | 2     | 3     | 4         | 5            |
|            |   | Impact        |       |       |           |              |

| Risk level |        |
|------------|--------|
|            | High   |
|            | Medium |
|            | Low    |

### 4.3 Optimising risk analysis

Risk analysis focuses on three major factors:

- **Risk assessment** involves identifying hazards to ensure the health and safety of personnel.
- **Risk management** awards those risks with mitigation strategies in order to plan the control and reduction of the impact or likelihood of said risk.

- **Risk communication** is the act of disseminating risks to stakeholders.

With regards to HALO REACH, there is a colossal number of risks that can result in the UAV missing its requirements. These can be identified in the models outlined in the Systems Engineering section. Therefore, it is of utmost importance that plans are put in place with relevant alleviation strategies for risks that pose a serious threat to the company or product.

#### 4.4 Risk categories

It is indispensable to explore all categories of risk, in order to prevent as many opportunities of failure as possible during the conceptual design phase. This assists with any probabilities of added cost during the lifecycle. A full breakdown of all risks considered can be seen in Appendix D.

##### 4.4.1 Regulatory compliance

Regulatory compliance risks result from non-compliance in regulations, standards and agreements set as a minimum requirement by relevant organisations, as listed in Section 2.3. These risks are normally mitigated against during the initial conceptualisation and design of the product. However, improper practices during the lifecycle by any party, whether this be from inadequate understanding or a lack of up-to-date regulatory monitoring, will result in operational disruptions. Therefore it is imperative that these risks are noticed and regularly monitored, as hampering the availability of REACH to any region or market will result in lack of fulfillment in REACH’s designated missions, costing money and lives.

One particular example of products not adhering to regulatory compliance is the 737 MAX [[30]. In 2018 and 2019, two 737 MAX aircraft were involved in complete hull losses and 346 deaths [31][32]. Investigations carried out by the FAA resulted in the discovery that Boeing withheld information about their Maneuvering Characteristics Augmentation System (MCAS) to their pilots, as well as the FAA. This directly violated the FAA 14 CFR Part 121, which details requirements for crew training and operational safety procedures [33]. Boeing has since come under intense scrutiny for its direct involvement in issuing airworthiness certificates for their 737 MAX aircraft [source FAA wikipedia article]. As illustrated in Section 2.3, HALO must ensure that all REACH UAV’s adhere to all relevant regulations, in order to sustain long-term operational success and keep stakeholder trust. We don’t want any potential whistleblower issues (Boeing, don’t @ me) [34]

| Risk ID | Risk Event                                | Risk description                                                                                                              | Root causes                                                                           | Effects                                                     |
|---------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------|
| RC01    | Failure to comply with regulations        | Includes lack of permits, non-compliance with airspace regulations, and violation of safety standards or industry guidelines. | Lack of awareness of regulations, inadequate regulatory oversight, poor documentation | Legal penalties, operational shutdowns, loss of credibility |
| RC02    | Legal liabilities and unauthorized access | Legal issues from accidents or incidents, and unauthorized access to the drone’s control system or data.                      | Poor security protocols, lack of insurance, inadequate access controls                | Financial losses, reputational damage, legal actions        |

#### 4.5 Human factors

The CAA states that human error is identified as ”the main cause or contributory factor in approximately 75% of all aviation accidents and incidents” [35]. Human factors need to be taken into account across every aspect of a system from analysis, design, development, implementation and the operation



use. It is essential to produce safe, comfortable and effective human performance [36]. Human actors range across three primary groups:

- **The job** encompasses the overall working environment. Each individual is different; we all have our own strengths and weaknesses. It is the task of the organisation to allot operations in a sensible and ordered manner, taking into account physical and mental capabilities of employees.
- **The individual** encloses the competence, personality, skills and attitudes of any involved person. [37]
- **The organisation** refers to work patterns, work culture and general resources. It is very easy to overlook key factors that may have significant effects on employees. [lecture notes 9]

Human factors have a direct relation to cost related risks, as it takes into account manpower, personnel and training. With regards to HALO, this could be in the form of environmental costs, operational costs, healthcare costs for any operators or general redesign costs over the lifespan of REACH. As REACH deals directly with the rescue operations of people, it is indispensable to consider human factors in all aspects of the HALO and REACH's behaviour. Numerous incidents and accidents have arisen due to the mismanagement of personnel or the UAV itself. Human factors and actions must be followed and monitored in order to prevent incidents in REACH's mission, such as following checklists and having multiple personnel monitoring the progression of missions.

| Risk ID | Risk Event                              | Risk description                                                                             | Root causes                                                         | Effects                                                                              |
|---------|-----------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| HF01    | Operator errors and inadequate training | Errors or negligence by drone operators, and inadequate training or experience of personnel. | Insufficient training programs, lack of experience, high turnover   | Increased accidents, operational inefficiency, higher maintenance costs              |
| HF02    | Coordination and communication issues   | Lack of coordination or communication between drone operators and other relief agencies.     | Lack of standardized protocols, insufficient communication channels | Delays in operations, misaligned efforts, reduced effectiveness of relief operations |

## 4.6 Technical

Technical risks involve challenges and uncertainties arising from the use and dependency of technological subsystems within the overall system of REACH. These risks can stem from failures of subsystem hardware, software or cybersecurity attacks. Managing these technical risks must therefore take into account regular software and hardware updates, improved monitoring of IT systems, as well as company-wide policies to educate employees about key technical aspect of REACH.

| Risk ID | Risk Event                     | Risk description                                                                          | Root causes                                                            | Effects                                                               |
|---------|--------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------|
| TC01    | Hardware and software failures | Malfunctions in critical components and software bugs affecting control system stability. | Poor quality control, inadequate testing, use of unreliable components | System downtime, mission failures, increased repair costs             |
| TC02    | Cybersecurity vulnerabilities  | Exposure to hacking or cyberattacks due to vulnerabilities in cybersecurity defenses.     | Outdated software, weak passwords, lack of security awareness          | Data breaches, loss of sensitive information, operational disruptions |

## 4.7 Environmental

These risks refer to threats posed by either natural or man-made factors which, for our drone and mission profiles, is a crucial factor in the operation of REACH. The drone must be able to operate in numerous harsh environments in order to provide disaster relief or communications relay. Tsunamis, storms, intense heatwaves or acidic conditions are only some of the settings that REACH must pilot through to achieve its mission requirements. Organisations specific to environmental sustainability and operations mitigate harsh environmental factors with rigorous assessments of conditions, compliance with local and international environmental condition and sustainable practices.

The drone itself is hybrid powered, which further enforces the requirement for an environmentally safe and sound drone in its operational environments. Entering harsh environments and leaving them for worse due to pollution can result in negative opinion from stakeholders, primarily in its secondary markets such as agriculture or industrial applications.

| Risk ID | Risk Event                       | Risk description                                                                                                   | Root causes                                                            | Effects                                                                     |
|---------|----------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| EN01    | Adverse environmental conditions | High winds, electromagnetic interference, or hazardous materials in the disaster area affecting drone performance. | Lack of environmental monitoring, poor planning for adverse conditions | Reduced drone performance, increased risk of accidents, mission failures    |
| EN02    | Collisions and wildlife impact   | Collisions with other aircraft, obstacles, and impact from wildlife (e.g., birds).                                 | Inadequate navigation systems, insufficient environmental awareness    | Damage to drones, increased maintenance costs, potential injury to wildlife |

## 4.8 Operational

Operational risks encompass disruptions and concerns within a company's affairs. These can result from controllable factors, such as production line inefficiencies, delays and equipment failure or from external factors, such as geopolitical instability and market volatility. Assessment and management of operational risks can be trickier due to external factors being unpredictable in most cases, which is exactly what REACH is designed to operate in. Eliminating these risks is extremely difficult to near impossible, hence sharing and undergoing risk financing is one of the best mitigation strategies for external operational risks.

| Risk ID | Risk Event                         | Risk description                                                                                              | Root causes                                                                | Effects                                                                  |
|---------|------------------------------------|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------|
| OC01    | Interference and navigation issues | Radio signal interference, loss of GPS signal, and difficulty navigating through tight or congested airspace. | Poor quality navigation systems, inadequate signal interference management | Loss of control, increased risk of accidents, operational inefficiencies |
| OC02    | Navigational error                 | Loss of GPS signal leading to navigation errors or loss of situational awareness .                            | Sensor malfunction, inadequate range calculation, terrain, signal loss     | Reduced operational range, inability to deliver necessary supplies       |

## 4.9 Cost

Cost risks can make or break a company; any mismanagement of funding, whether it be budgeting, general expenditure or investment can result in anything from loss of stakeholders and investors to bankruptcy and the cease of operations. Risks of this type can arise from any given area, from cost overruns, cost estimation errors and general mismanagement. As HALO requires an intense initial investment to procure manufacturing equipment, research, materials and other components for the inception of REACH, management of these funds is crucial. People will rely on REACH to provide them with immediate relief and communication, therefore this should be the driving factor in ensuring no financial fraud or mismanagement occurs, as well as sticking as close to timelines and budgets as possible.

Cost risks are found across all industries, however the UAV sector is guilty of unreasonable expenditures and lack of funding management. The F-35 faces numerous production and supply chain issues over its lifecycle and refurbishment stages, with regular modernisation efforts resulting in a vast increase in its budget requirement [38]. Any hiccups in the supply chain result in a point of failure for the aircraft [39]. It is therefore pivotal that HALO sticks to the outlined budgets as best as we can, diversify our suppliers to avoid late supply chain issues and have a good working relationship with suppliers to further mitigate against avoidable cost risks.

| Risk ID | Risk Event                            | Risk description                                                                                               | Root causes                                                           | Effects                                           |
|---------|---------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------|
| CO01    | Dependency on a single funding source | Vulnerability due to reliance on one funding source.                                                           | Lack of diversified funding strategies, over-reliance on one source   | Financial instability, project disruptions        |
| CO02    | Underestimate of costs                | Underestimation of development, production, and operational deployment costs, resulting in budget constraints. | Inaccurate budgeting, lack of historical data, optimistic projections | Budget overruns, project delays, financial strain |

## 4.10 Risk management and mitigation strategies

As illustrated in several of the above examples, it is a requirement to manage and mitigate risks in all aspects of the company and product operation. There are general steps one should follow in successfully managing risks within a project:

- **Identify** potential risks
- **Analyse** and **assess** risks
- **Develop** a risk plan
- **Execute** the risk plan
- **Track and monitor** the risks.

Following this universal schedule allows for one to determine what resources to allocate to each risk, as well as monitoring the development and mitigation process of said risk. With regards to all risks within HALO REACH, the overall goal is to reduce them to their lowest possible level, ensuring successful and long lasting operation of REACH.

ISO 31000:2009 gives a list on several effective mitigation techniques for risks that will be prevalent in HALO REACH: **avoiding** the risk by eliminating the risky activity, **removing** the risk source completely and **changing** the consequences or impact of the risk are just some of the mitigating techniques underlined and pursued by groups and projects worldwide. With further operation and testing of HALO REACH, reliability analysis can be introduced. This will greatly aid the risk mitigation process by analysing probabilities of failure for individual components to whole subsystems. This will allow for designing REACH for redundancy in all subsystems and components, boosting overall efficiency and functionality of REACH.

A table of risks which need to be mitigated for, due to their risk matrix ranking, can be found in Appendix ??

## 5 Conclusion

The UAS for Disaster Relief posed a very interesting and intense 5 weeks of meticulous management, in order to generate a UAS which could fulfill the mission requirement to the best of our capabilities. The importance of people management and person skills when project managing on any scale is incredibly important, in order to further enhance team cohesion and progress. Giving every person an equal say with regards to the mission architecture and project route allows for a healthy, enjoyable workspace, promoting an active culture where people can freely communicate their concerns and ideas, whilst advancing with their respective subgroup. This was made possible by following the systems engineering skills outlined through this report, as well as regular engagement with all parties involved. The risk analysis posed difficult with regards to quantifying potential problems and hazards that could arise at any given point during the overall system life cycle, despite the group only performing the initial concept stages. To ensure further success in this project, further exploration of risk prevention through redundancy must occur, alongside the development, validation and verification of system components and subsystems. Overall, HALO REACH can prove to be a sustainable, competitive figure in this new, undeveloped market. With climate disasters and insurgency seeming to never dissipate, it is crucial that endeavors such as this are explored and pursued thoroughly to benefit society as a whole.

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# Appendix

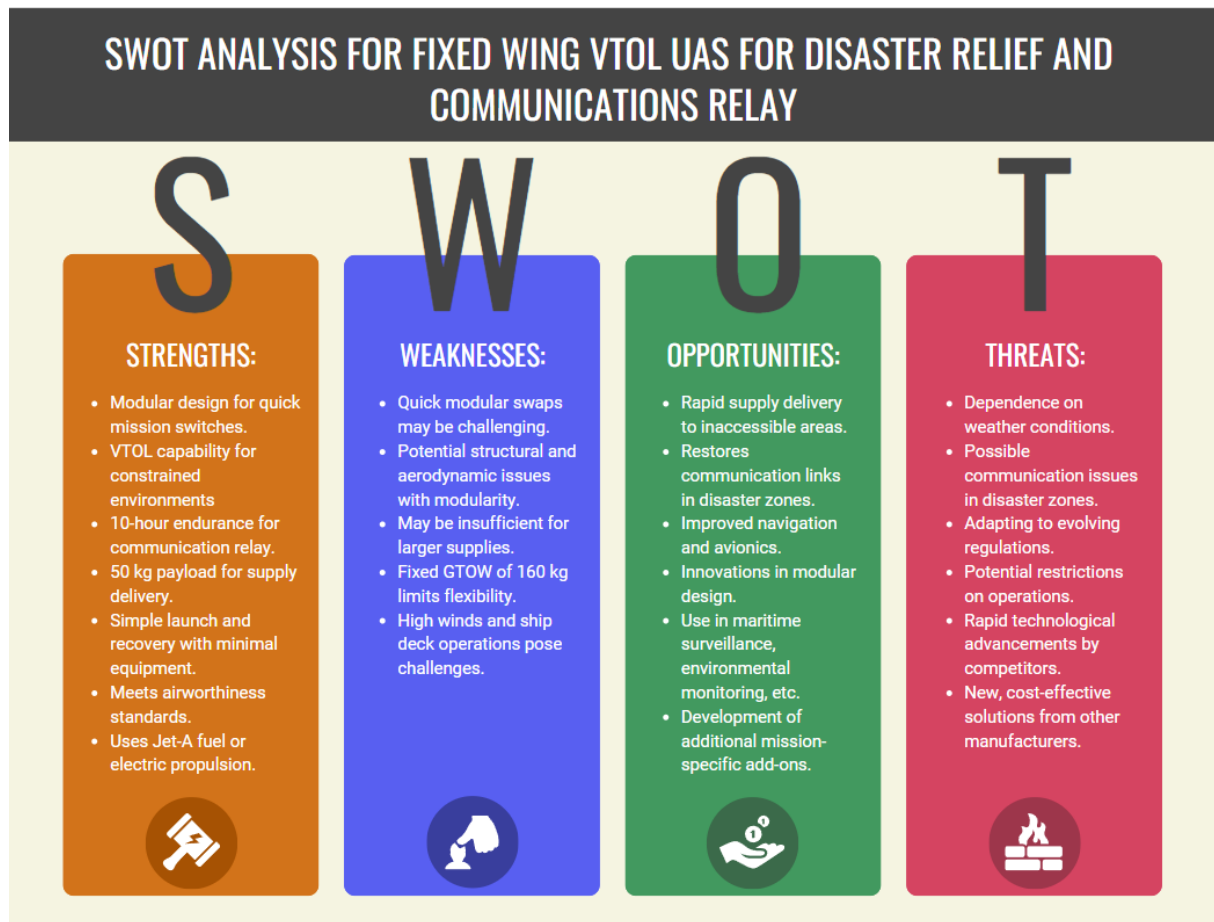
## A N<sup>2</sup> chart

|              |                        |                    |                         |                            |                               |                           |                |     |           |                         |                                   |
|--------------|------------------------|--------------------|-------------------------|----------------------------|-------------------------------|---------------------------|----------------|-----|-----------|-------------------------|-----------------------------------|
| Rotor design |                        | x                  | x                       |                            | x                             |                           | x              | x   |           |                         |                                   |
|              | Lifting surface design | x                  |                         | x                          | x                             | x                         | x              | x   | x         | x                       | x                                 |
|              | x                      | Aerofoil selection |                         | x                          | x                             |                           | x              |     |           |                         |                                   |
| x            |                        |                    | Rotor structural design | x                          | x                             | x                         | x              |     |           |                         |                                   |
| x            | x                      |                    | x                       | Aircraft structural layout | x                             | x                         | x              |     | x         | x                       | x                                 |
|              |                        |                    |                         |                            | Material/manufacturing method |                           | x              |     |           |                         |                                   |
|              |                        |                    |                         |                            | x                             | Powerplant type selection | x              |     | x         |                         |                                   |
|              |                        |                    |                         | x                          | x                             |                           | Weight/Balance |     | x         | x                       | x                                 |
|              |                        |                    |                         |                            | x                             |                           |                | GNC |           |                         |                                   |
|              |                        |                    |                         |                            | x                             |                           |                |     | FC design | x                       | x                                 |
|              |                        |                    |                         |                            | x                             |                           |                | x   |           | Add-on subsystem design | x                                 |
|              |                        |                    |                         |                            | x                             |                           |                |     |           | x                       | Add-on aerodynamic considerations |

**Figure 5.** Caption



## B SWOT analysis



**Figure 6.** SWOT analysis performed for HALO REACH

## C House of Quality

| Customer Requirements                  |     |           |         |     |                        |             |            |         |                                     |
|----------------------------------------|-----|-----------|---------|-----|------------------------|-------------|------------|---------|-------------------------------------|
| Cost-effectiveness                     | 5   | 9         | 9       | 3   | 3                      | 9           | 3          | 3       | 9                                   |
| High Reliability                       | 5   | 9         | 9       | 3   | 9                      | 9           | 3          | 9       | 9                                   |
| Low Maintenance Requirements           | 4   | 9         | 3       | 3   | 3                      | 9           | 9          | 1       | 3                                   |
| Scalability                            | 4   | 3         | 9       | 3   | 3                      | 3           | 3          | 1       | 3                                   |
| Portability                            | 3   | 3         | 9       | 3   | 3                      | 3           | 3          | 3       | 3                                   |
| Extended Operational Range             | 5   | 3         | 9       | 9   | 9                      | 9           | 9          | 9       | 3                                   |
| Environmental Sustainability           | 4   | 3         | 9       | 3   | 3                      | 3           | 3          | 3       | 3                                   |
| Low Noise Operation                    | 4   | 3         | 3       | 3   | 3                      | 3           | 3          | 3       | 3                                   |
| Data Security                          | 4   | 1         | 3       | 3   | 1                      | 3           | 3          | 1       | 1                                   |
| Interoperability with Existing Systems | 4   | 3         | 3       | 3   | 3                      | 3           | 3          | 3       | 3                                   |
| Importance rating                      | 217 | 231       | 139     | 237 | 193                    | 147         | 191        | 223     | 163                                 |
| WingtraOne GEN II                      | No  | 1 hour    | 0.8kg   | No  | 18m/s gust resistance  | Easy        | Global     | Battery | 10km radio LOS                      |
| Trinity F90+                           | Yes | 1.5 hours | 2kg     | Yes | 18m/s gust resistance  | Medium      | EASA       | Battery | 7.5km radio LOS                     |
| SMD V800                               | Yes | 10 hours  | 50kg    | Yes | 14m/s gust resistance  | Medium      | Only China | Hybrid  | 100km radio with GCS                |
| HALO REACH                             | Yes | 10 hours  | 20/50kg | Yes | >18m/s gust resistance | Easy/Medium | Global     | Hybrid  | 185km transmitter/receiver with GCS |
| Competitive evaluation                 |     |           |         |     |                        |             |            |         |                                     |
| Targets                                |     |           |         |     |                        |             |            |         |                                     |
| Customer importance rating             |     |           |         |     |                        |             |            |         |                                     |
| Vertical Takeoff and Landing           |     |           |         |     |                        |             |            |         |                                     |
| Endurance                              |     |           |         |     |                        |             |            |         |                                     |
| Payload Capacity                       |     |           |         |     |                        |             |            |         |                                     |
| Modular Design                         |     |           |         |     |                        |             |            |         |                                     |
| Weather Resistance                     |     |           |         |     |                        |             |            |         |                                     |
| Ease of Use                            |     |           |         |     |                        |             |            |         |                                     |
| Compliance                             |     |           |         |     |                        |             |            |         |                                     |
| Fuel Compatibility                     |     |           |         |     |                        |             |            |         |                                     |
| Gross Takeoff Weight                   |     |           |         |     |                        |             |            |         |                                     |
| Communications System                  |     |           |         |     |                        |             |            |         |                                     |
| Competitor Research                    |     |           |         |     |                        |             |            |         |                                     |
| WingtraOne GEN II                      |     |           |         |     |                        |             |            |         |                                     |
| Trinity F90+                           |     |           |         |     |                        |             |            |         |                                     |
| SMD V800                               |     |           |         |     |                        |             |            |         |                                     |

Figure 7. House of Quality analysis of REACH

D Complete risk table

| Risk ID                                                    | Risk Event                                   | Risk description                                                                                                                                         | Root causes                                                                                        | Effects                                                                                             | Time | Impact analysis |   | Quality | Qualitative analysis |   | Confidence level% | Risk score | Risk level      | Response strategy |
|------------------------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|------|-----------------|---|---------|----------------------|---|-------------------|------------|-----------------|-------------------|
| Regulatory Compliance                                      | Failure to comply with regulations           | Includes lack of permits, non-compliance with industry regulations, and violations of safety standards or industry guidelines.                           | Lack of awareness of regulations, inadequate regulatory oversight, poor documentation              | Legal penalties, operational shutdowns, loss of credibility                                         | 8    | 9               | 7 | 9       | 8                    | 4 | 30%               | 32         | SOFT            | AVOID             |
|                                                            | Legal liabilities and unauthorized access    | Legal issues from accidents or incidents, and unauthorized access to the device's internal systems or data.                                              | Poor security protocols, lack of insurance, inadequate access controls                             | Financial losses, reputational damage, legal actions                                                | 7    | 8               | 6 | 9       | 8                    | 5 | 40%               | 40         | SOFT            | AVOID             |
|                                                            | Operator errors and inadequate training      | Errors or negligence by device operators during operations or emergency response.                                                                        | Inefficient training programs, lack of experience, high turnover                                   | Increased accidents, operational inefficiency, higher maintenance costs                             | 6    | 7               | 6 | 8       | 7                    | 6 | 50%               | 42         | MEDIUM          | MITIGATE          |
| Human Factors                                              | Coordination and communication issues        | Lack of coordination or communication between operators and other related agencies.                                                                      | Lack of standardized protocols, insufficient communication channels                                | Delays in operations, misaligned efforts, reduced effectiveness of relief operations                | 8    | 7               | 8 | 7       | 8                    | 7 | 45%               | 56         | MEDIUM          | MITIGATE/AVOID    |
|                                                            | Hardware and software failures               | Malfunctions in critical components such as sensors, propulsion systems, or communication devices, and software bugs affecting overall system stability. | Poor quality control, inadequate testing, use of unreliable components                             | System downtime, mission failures, increased repair costs                                           | 9    | 9               | 7 | 9       | 9                    | 8 | 55%               | 72         | HARD            | AVOID             |
| Technology Risks                                           | Cybersecurity vulnerabilities                | Exposure to hacking or cyberattacks due to vulnerabilities in cybersecurity defenses.                                                                    | Outdated software, weak passwords, lack of security awareness                                      | Data breaches, loss of sensitive information, operational disruptions                               | 7    | 8               | 7 | 9       | 8                    | 6 | 50%               | 48         | MEDIUM          | TRANSFER          |
|                                                            | Adverse environmental conditions             | High winds, electromagnetic interference, or hazardous materials in the disaster area affecting device performance.                                      | Lack of environmental monitoring, poor planning for adverse conditions                             | Reduced device performance, increased risk of accidents, potential damage to device                 | 7    | 6               | 7 | 7       | 8                    | 7 | 40%               | 56         | MEDIUM          | MITIGATE          |
| Environmental Hazards                                      | Collisions and debris impact                 | Collisions with other aircraft, obstacles, and impact from falling debris.                                                                               | Inadequate navigation systems, inadequate environmental awareness, lack of common sense technology | Damage to device, increased maintenance costs, potential damage to wildlife                         | 8    | 8               | 7 | 8       | 8                    | 6 | 45%               | 48         | MEDIUM          | MITIGATE/ACCEPT   |
|                                                            | Interference and navigation issues           | Radio signal interference, loss of GPS signal, and difficulty navigating through tight corridors or congested areas.                                     | Poor quality navigation system, inadequate signal strength, poor weather conditions                | Loss of control, increased risk of accidents, operational inefficiencies                            | 7    | 7               | 6 | 8       | 8                    | 7 | 50%               | 56         | MEDIUM          | MITIGATE/AVOID    |
| Operational Constraints                                    | Interference and navigation issues           | Radio signal interference, loss of GPS signal, and difficulty navigating through tight corridors or congested areas.                                     | Poor quality navigation system, inadequate signal strength, poor weather conditions                | Loss of control, increased risk of accidents, operational inefficiencies                            | 7    | 7               | 6 | 8       | 8                    | 7 | 50%               | 56         | MEDIUM          | MITIGATE/AVOID    |
|                                                            | Battery life and payload capacity            | Limited battery life affecting operational range and duration, and insufficient payload capacity for delivering essential supplies.                      | Poor battery technology, inadequate energy management, lack of payload optimization                | Reduced operational range, inability to deliver necessary supplies, mission failures                | 8    | 7               | 8 | 7       | 8                    | 7 | 55%               | 56         | MEDIUM          | MITIGATE/AVOID    |
| I & D - Conceptual Design Ideas                            | Lack of thorough research and innovation     | Failure to conduct thorough research on existing designs, leading to missed opportunities for innovation.                                                | Insufficient investment in R&D, lack of collaboration with research institutions                   | Missed innovation opportunities, falling behind competitors, inefficiencies in design               | 6    | 7               | 8 | 7       | 6                    | 6 | 35%               | 36         | SOFT            | MITIGATE          |
|                                                            | Inadequate interdisciplinary collaboration   | Lack of collaboration among different design disciplines or engineering specialties.                                                                     | Siloed teams, lack of integrated project management                                                | Design flaws, overlooked issues, reduced overall effectiveness                                      | 7    | 6               | 7 | 7       | 7                    | 7 | 40%               | 42         | MEDIUM          | AVOID             |
|                                                            | Inadequate demand assessment                 | Failure to accurately assess demand for disaster relief devices, leading to overproduction or underutilization.                                          | Lack of market research, poor data analysis methods                                                | Overproduction or underutilization of devices, financial losses, inefficiencies                     | 7    | 7               | 8 | 6       | 7                    | 6 | 45%               | 42         | SOFT            | MITIGATE/ACCEPT   |
| Market Research                                            | Regulatory and logistical challenges         | Inadequate understanding of regulatory requirements, leading to delays in deployment.                                                                    | Inadequate understanding of regulatory environments, poor communication with regulators            | Delays in deployment, increased costs, operational inefficiency                                     | 8    | 7               | 8 | 7       | 8                    | 7 | 50%               | 56         | MEDIUM          | MITIGATE/AVOID    |
|                                                            | Dependence on volatile funding               | Reliance on unstable funding sources (e.g., grants, donations) without sustainable revenue streams.                                                      | Reliance on grants or donations, lack of sustainable revenue streams                               | Project disruptions, inability to scale, financial instability                                      | 8    | 8               | 8 | 7       | 8                    | 6 | 45%               | 48         | MEDIUM          | AVOID             |
| Gather Budget                                              | Underestimation of costs                     | Underestimation of development, production, and operational deployment costs.                                                                            | Inaccurate budgeting, lack of historical data, optimistic projections                              | Budget overruns, project delays, financial strain                                                   | 7    | 9               | 8 | 7       | 8                    | 6 | 50%               | 48         | MEDIUM          | MITIGATE          |
|                                                            | Dependence on volatile funding               | Reliance on unstable funding sources (e.g., grants, donations) without sustainable revenue streams.                                                      | Reliance on grants or donations, lack of sustainable revenue streams                               | Project disruptions, inability to scale, financial instability                                      | 8    | 8               | 8 | 7       | 8                    | 6 | 45%               | 48         | MEDIUM          | AVOID             |
| Internal Stakeholders                                      | Lack of alignment and communication          | Misalignment among teams regarding project goals and priorities, and inadequate communication channels.                                                  | Poor leadership, lack of clear goals, ineffective communication channels                           | Misaligned project goals, reduced team morale, inefficiency in operations                           | 7    | 6               | 7 | 7       | 6                    | 6 | 40%               | 36         | SOFT            | MITIGATE/ACCEPT   |
|                                                            | Failure to address internal concerns         | Delayed or ineffective response to concerns raised by internal stakeholders, leading to frustration.                                                     | Inadequate feedback mechanisms, poor internal communication                                        | Reduced employee morale, unresolved issues, reduced productivity                                    | 6    | 6               | 7 | 6       | 6                    | 5 | 35%               | 30         | SOFT            | MITIGATE/ACCEPT   |
| External Stakeholders                                      | Inefficient engagement and collaboration     | Limited engagement with external partners, leading to missed opportunities for resources or expertise.                                                   | Limited outreach efforts, lack of partnership strategies                                           | Reduced resources, operational inefficiencies, missed opportunities for collaboration               | 7    | 6               | 7 | 6       | 7                    | 6 | 40%               | 42         | SOFT            | MITIGATE          |
|                                                            | Lack of transparency and accountability      | Poor communication with external stakeholders, leading to trust and credibility issues.                                                                  | Poor governance structures, lack of open communication                                             | Erosion of trust, operational damage, loss of stakeholder confidence                                | 7    | 7               | 7 | 8       | 8                    | 4 | 45%               | 32         | MITIGATE/ACCEPT |                   |
| Funding                                                    | Dependency on a single funding source        | Vulnerability due to reliance on one funding source.                                                                                                     | Lack of diversified funding strategies, over-reliance on one source                                | Financial instability, project disruptions, inability to scale                                      | 8    | 8               | 8 | 7       | 8                    | 6 | 50%               | 48         | MEDIUM          | AVOID             |
|                                                            | Mismanagement of funds                       | Inefficiencies or allegations of financial misappropriation due to mismanagement or misuse of funds.                                                     | Lack of financial controls, inadequate oversight, poor financial planning                          | Financial losses, legal issues, reputational damage                                                 | 8    | 9               | 7 | 8       | 9                    | 6 | 55%               | 54         | MEDIUM          | AVOID             |
| Key Design Characteristics                                 | Compromising on quality or performance       | Sacrificing design quality or performance standards to meet cost or time constraints.                                                                    | Pressure to reduce costs, inadequate quality assurance processes                                   | Poor performance of devices, increased maintenance costs, reduced reliability                       | 7    | 7               | 7 | 9       | 8                    | 5 | 50%               | 40         | SOFT            | MITIGATE/AVOID    |
|                                                            | Inadequate testing and validation            | Insufficient testing of design concepts or prototypes, leading to unforeseen issues.                                                                     | Insufficient testing protocols, rushed development timelines                                       | Recurring issues, inefficiencies in development, increased maintenance costs, mission failures      | 8    | 8               | 7 | 9       | 9                    | 6 | 55%               | 54         | HARD            | AVOID             |
| Technical Specification                                    | Incomplete documentation and recurring flaws | Inaccurate documentation of specifications and failure to learn from previous projects.                                                                  | Poor documentation practices, lack of continuous improvement processes                             | Unresolved issues, inefficiencies in development, increased maintenance costs, mission failures     | 7    | 7               | 6 | 8       | 8                    | 6 | 50%               | 48         | MEDIUM          | AVOID             |
|                                                            | Lack of scalability or interoperability      | Hardware is future upgrade or integration with other technologies.                                                                                       | Proprietary systems, lack of standardization                                                       | Difficulty to integrate new technologies, limitations in expansion, reduced flexibility             | 7    | 7               | 8 | 7       | 7                    | 6 | 45%               | 42         | SOFT            | MITIGATE          |
| Simulate/Verify Structural Response and Component Function | Insufficient real-world testing              | Inadequate simulation or verification of structural response under real-world conditions.                                                                | Over-reliance on simulations, lack of field testing                                                | Unanticipated failures, reduced effectiveness in real world conditions, increased operational risks | 8    | 8               | 7 | 9       | 9                    | 7 | 55%               | 63         | HARD            | AVOID             |
|                                                            | Lack of collaboration                        | Disconnects between simulation and performance due to poor coordination.                                                                                 | Siloed teams, lack of integrated project management                                                | Integration failures, operational inefficiencies, increased costs                                   | 7    | 6               | 7 | 7       | 8                    | 6 | 40%               | 42         | SOFT            | MITIGATE          |
| Systems/Digital Integration                                | Interoperability issues                      | Challenges in integrating different systems or digital components.                                                                                       | Use of incompatible components, lack of standardization                                            | Data transfer issues, operational inefficiencies, increased costs                                   | 7    | 7               | 7 | 8       | 8                    | 6 | 50%               | 48         | MEDIUM          | AVOID/TRANSFER    |
|                                                            | Cybersecurity risks                          | Vulnerabilities exposing the device system to potential hacking or data breaches.                                                                        | Outdated software, weak passwords, lack of security protocols                                      | Data breaches, operational disruptions, loss of sensitive information                               | 7    | 8               | 7 | 9       | 8                    | 7 | 50%               | 56         | HARD            | AVOID/TRANSFER    |
| Product Manufacturing and Distribution                     | Unreliable manufacturers and quality control | Selection of unreliable manufacturers and inadequate quality control measures.                                                                           | High failure rates, inadequate quality control processes                                           | High failure rates, increased maintenance costs, operational disruptions                            | 8    | 8               | 7 | 8       | 8                    | 7 | 50%               | 56         | HARD            | AVOID             |
|                                                            | Distribution delays                          | Disruptions in the distribution or delivery of device systems, affecting disaster relief operations.                                                     | Poor logistical planning, unreliable shipping partners                                             | Delays in relief operations, increased costs, reduced effectiveness                                 | 8    | 7               | 7 | 7       | 7                    | 6 | 45%               | 48         | MEDIUM          | AVOID             |
| Product Promotion and Customer Support                     | Ineffective marketing strategies             | Limited awareness or adoption due to poor marketing.                                                                                                     | Lack of market research, poor branding and promotional efforts                                     | Limited awareness, low adoption rates, reduced sales                                                | 7    | 6               | 8 | 6       | 6                    | 6 | 40%               | 36         | SOFT            | MITIGATE/ACCEPT   |
|                                                            | Inadequate customer support                  | Insufficient customer support leading to user frustration and dissatisfaction.                                                                           | Insufficient support infrastructure, lack of training for support staff                            | User frustration, reduced customer satisfaction, damage to reputation                               | 6    | 6               | 7 | 7       | 6                    | 4 | 40%               | 24         | SOFT            | MITIGATE/ACCEPT   |

Figure 8. Complete risk table including all details for most important detailed risks for each section

## E Mitigation techniques

**Table 4.** List of relevant mitigation techniques for most intense risks

| <b>Risk ID</b> | <b>Risk Event</b>                     | <b>Original risk score</b> | <b>Mitigation technique</b>                           | <b>New risk score</b> |
|----------------|---------------------------------------|----------------------------|-------------------------------------------------------|-----------------------|
| HF01           | Hardware and software failures        | 5B                         | Redundancy and failover systems                       | 3B                    |
| TR01           | Cybersecurity vulnerabilities         | 5B                         | Encryption, firewalls                                 | 3A                    |
| EN01           | Adverse environmental conditions      | 5D                         | Regular weather forecast and monitoring               | 3D                    |
| OC01           | Interference and navigation issues    | 5A                         | Anti-jamming technologies                             | 3A                    |
| OC02           | Navigational error                    | 5B                         | Advanced navigation algorithms and mapping techniques | 3A                    |
| CO01           | Dependency on a single funding source | 4C                         | Diversify funding sources                             | 2B                    |